

TUTORIAL 01 – THE RHINO CASE

Scenario

The city of New Orleans passed a law in 2004 making possession of nine or more unique rhinoceros images a serious crime. The network administrator at the University of New Orleans recently alerted police when his instance of RHINOVORE flagged illegal rhino traffic. Evidence in the case includes a computer and USB key seized from one of the University's labs.

Unfortunately, the computer had no hard drive. The USB key was imaged and a copy of the dd image can be obtained from: https://cfredsarchive.nist.gov/dfrws/Rhino_Hunt.html.

In addition to the USB key drive image, three network traces are also available—these were provided by the network administrator and involve the machine with the missing hard drive. The suspect is the primary user of this machine, who has been pursuing his Ph.D. at the University since 1972.

MD5 hashes for evidence:

- c0d0093eb1664cd7b73f3a5225ae3f30 – for rhino.log
- cd21eaf4acfb50f71ffff857d7968341 - rhino2.log
- 7e29f9d67346df25faaf18efcd95fc30 - rhino3.log
- 80348c58eec4c328ef1f7709adc56a54 - RHINOUSB.dd

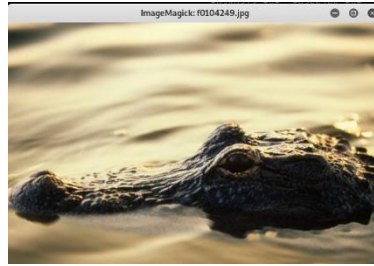
PART ONE: ANALYSIS WITH AUTOPSY

Create a new case – Rhino_Tutorial_Part_One, and upload the content of the USB image. Analyse the content of the image and find proof (images of rhinos) that, the owner of this key committed a crime.

- Verify that the USB image and log files have not been tempered with.**
- What are \$CarvedFiles and \$OrphanFiles?**
- What was the file system of this USB key?**
- Who gave the accused a telnet/ftp account?**
- What other useful hints did you get from analysing the files in the USB image – like email address, what happened to the PC hard drive, who created him a telnet/ftp account?**

PART TWO: STEGANOGRAPHY

Still using the same USB image, extract the deleted crocodile image and alligator from the data image using the autopsy tool (look like this):



Step 3: Using the following tools in linux:

- Stegdetect
- steghide
- stegseek
- jphide
- jpseek
- jpsh_bruteforce
- stegbreak
- stegcracker

To detect, hack the passkey (used for steganography) and extract images of rhinos hidden in the alligator and crocodile images.

- a) What were the passwords, used for the hiding and extracting the images in the alligator and crocodile images?
- b) Show us the extracted (hidden) rhino images.

PART THREE: NETWORK LOG ANALYSIS

Open the “rhino.log” file for network analysis with Wireshark. First, we will analyze unencrypted data traffic sent over the network (if there is any) and try to extract any useful information necessary. In case the investigators need probable cause and extra details for a warrant, this analysis can provide us with extra information or evidence.

To analyze unencrypted data traffic, click on Tools > Credentials.

For each of the unencrypted data traffic discovered (if any), analyze the data and extract any relevant information for the investigation.

- a) Which protocol was used for downloading unencrypted traffic from the server?
- b) Reconstruct the hex data of the downloaded image, to retrieve the rhino image
- c) What was the credentials (username and password) used for connecting to the server?
- d) Where is this server hosted (location or other details)?